A provenance-tracked State-day panel of India's power system built from official public reports, 2018–2026

Rohit Kumar[1]

[1]Centre for Artificial Intelligence, Dr. B. R. Ambedkar National Institute of Technology Jalandhar, Jalandhar 144008, Punjab, India

ORCID: Rohit Kumar 0009-0004-7936-9683

Corresponding author: Rohit Kumar (therohitkumar.build@gmail.com)

Abstract

India publishes detailed daily information on electricity demand, scheduling, generation and prices, but as thousands of report files whose layouts change over time, spread across the grid operator, the Central Electricity Authority and the meteorological department. We describe a State-day panel for 34 Indian State control areas from 2018-04-01 to 2026-08-31 (104,550 State-days, 107 variables). It joins, from official sources only, energy met and shortages, scheduled and actual drawal with over- and under-drawal, conventional generation by fuel, renewable generation, bid-area exchange prices and deviation charge rates, system frequency stress, carbon intensity and weather. Every one of the 18,233 source files is recorded with its URL, retrieval time and SHA-256 hash, and the panel can be rebuilt deterministically from the raw files. Each variable carries its measured publication delay, so forecasting studies can respect what was known on each day. Technical validation reproduces the sources' own totals exactly and agrees with an independent copy of the generation data within 1% on 100.0% of 75,485 State-days.

Background & Summary

India operates one of the world's largest synchronous grids, and its States are both the units of demand and the units that are scheduled, settled and regulated. Each day the national load despatch centre of Grid Controller of India Limited (Grid-India) publishes a Power Supply Position report with State demand, scheduled and actual drawal and the resulting over- or under-drawal [1]. The Central Electricity Authority (CEA) publishes station-wise generation [2] and State renewable generation [3] and maintains the national CO_2 baseline database [4]. Grid-India publishes the bid-area exchange prices and deviation charge rates used for settlement [5] under successive deviation settlement regulations [6, 7]. The India Meteorological Department (IMD) distributes gridded daily rainfall and temperature [8–10].

These sources are public, but they are not a dataset. They are PDF and spreadsheet reports whose sheet order, column positions, date conventions and even file names change over the years. Several failure modes affect naive use:

- report pages that carry the previous day's date;
- two differently defined renewable tables in the same workbook, in varying order;
- a web archive that points many report dates to one shared file;
- typing errors in station records;
- sources that move to a new address or stop publishing without notice.

Studies of Indian electricity at State level have therefore relied on hand-collected extracts or third-party aggregations whose processing is not documented, which limits reproducibility and comparison.

This data descriptor documents a panel built for a research programme on forecasting, context-aware assessment of schedule deviations and risk-aware scheduling. It follows three principles.

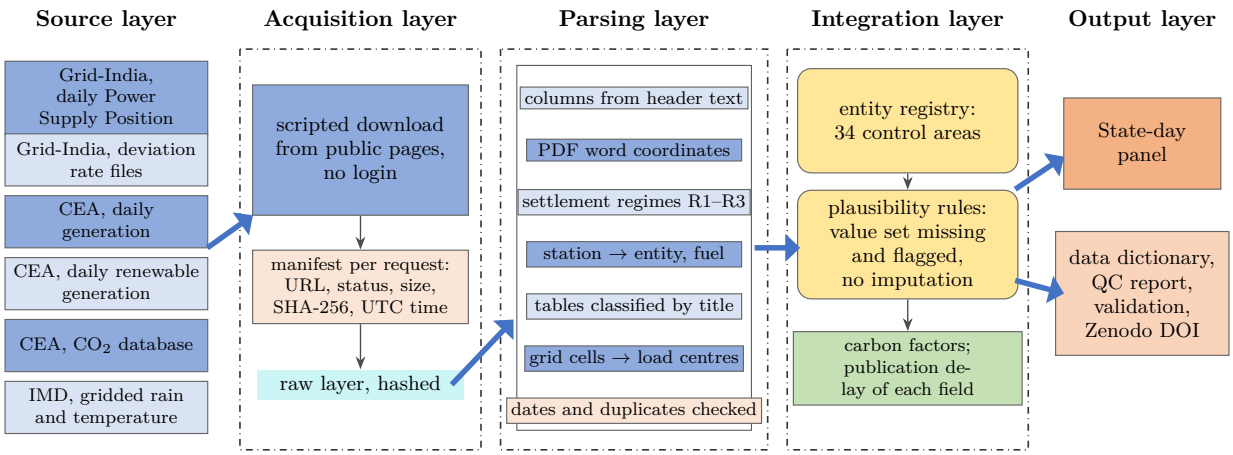


Figure 1: How the panel is built: official reports from six products are downloaded with a hashed record of every request, parsed by source-specific readers, mapped to 34 control areas and checked by plausibility rules; the release holds the panel, its data dictionary, the quality-control report and the validation. Each stage writes logs.

1. **Official sources only.** Every value comes from a report published by Grid-India, CEA or IMD. Aggregator copies are used solely as an external cross-check.
2. **Provenance and determinism.** Each raw file is stored unchanged with URL, retrieval time, size and SHA-256 hash; parsers and build steps are deterministic, and the panel carries a content checksum (`8099e9fdc72c7ea7...`).
3. **Information-set awareness.** Each variable is annotated with the delay after which it is publicly available, measured from the publishers' own file metadata, so that forecasting and decision studies do not use data that were not yet published.

The design follows the documentation practice of datasheets for datasets [11], the FAIR principles [12] and the processing transparency advocated for European power system data [13]. In Europe, a central transparency platform and processed open datasets of this kind exist [13, 14], together with reanalysis-based series of renewable output [15, 16] on which studies of renewable generation build [17], and open data and software are increasingly seen as a condition for reproducible energy research [18]. This descriptor documents a comparable panel for the State control areas of India, built from the official reports alone. Two companion manuscripts, on day-ahead forecasting and on day-ahead scheduling, use version 1.0 of the panel; neither had been published when this descriptor was written. This descriptor documents version 1.1, which adds the renewable generation reports that the Central Electricity Authority has published on a new portal since November 2025.

Figure 1 summarises the workflow. The panel (104,550 State-days × 107 columns, 10.25 million values) covers 34 State control areas and 3,075 days.

Methods

Sources and acquisition

Table 1 lists the sources. All files were downloaded without login from public pages by scripted HTTP requests on 14–15 September 2026, and the renewable reports published on CEA's new portal on 25 October 2026. Grid-India reports were enumerated through the file listing interface of its website, which returns one record per published attachment (4,834 records for the Power Supply Position reports). CEA National Power Portal reports follow a date-based URL pattern. CEA renewable reports up to 18 November 2025 were enumerated from the listing behind the report page of cea.nic.in. From November 2025 CEA publishes the same report on its RE genera-

Table 1: Official sources used in the panel. Files are those retrieved successfully; "not available" counts requests that returned an error or no report.

Publisher and product	Content used	Files	Not avail.	Size (MB)
Grid-India, daily Power Supply Position report	State demand, shortages, drawal schedule, over/under-drawal, max over/under-drawal; regional generation and inter-regional flows; frequency profile	3,448	5	2667.2
Grid-India, deviation settlement rate files	bid-area day-ahead and real-time clearing prices, normal rate of charges for deviation	404	0	87.6
CEA National Power Portal, daily generation report (sub-report 2)	station programme and actual generation, monitored capacity, capacity under outage, coal-stock days	3,022	74	1053.1
CEA, daily renewable generation report (cea.nic.in listing, to 18 Nov 2025)	State wind, solar and other renewable generation	2,361	8	382.3
CEA, daily renewable generation report (RE generation portal, from 1 Nov 2025)	as above	304	1	13.4
CEA, CO_2 baseline database (v22.0 and earlier)	station net generation and absolute emissions by financial year	48	0	68.0
IMD Pune, gridded daily data	rainfall (0.25°), maximum and minimum temperature (1°)	27	0	254.2

tion portal (`https://gen-re.cea.gov.in/reports`), one file per day under a date-based name; the old listing has no later report. Version 1.1 of the panel takes the portal files from 1 November 2025, and the old listing is kept on every date it covers. IMD gridded files are served per year and variable.

Every request, including those that failed, is written to a per-source manifest with URL, local path, HTTP status, size, SHA-256 hash and UTC retrieval time. A response smaller than a source-specific minimum size is recorded as unavailable rather than stored. During acquisition a local network login page intercepted some requests; such pages are detected by content and are never stored, and the affected requests were retried. The raw layer holds 18,233 files (5.03 GB). An earlier pilot collection of aggregator copies (162 files) is kept unchanged with its own hash manifest and is used only for cross-checks.

Entities

The unit of analysis is the State control area as reported by Grid-India. There are 34 entities: all 28 States; five Union-Territory control areas covering six of India's eight Union Territories, namely Delhi, Chandigarh, Puducherry, Dadra and Nagar Haveli and Daman and Diu (reported separately before their merger and summed into one entity), and Jammu and Kashmir together with Ladakh, which the reports give as one area; and the control area of the Damodar Valley Corporation, a central utility. The Andaman and Nicobar Islands and Lakshadweep do not appear in the reports. Industrial and railway entities that appear in some reports are excluded. A registry maps every spelling used by any source to an entity, and assigns each entity to its region and exchange bid area; no label in any parsed report remained unmapped.

Parsing and harmonisation by source

Power Supply Position reports. Spreadsheet reports are read by locating the State table and mapping columns from their header text; positional reading fails on files that contain empty spacer columns. PDF reports are read from word coordinates: lines are clustered by vertical position, and numbers are assigned to columns by their horizontal centres. The data date is the printed "Date of Reporting" minus one day.

In 2 cases, two reports with different figures carried the same printed date because the table page had not been updated. The report whose cover letter implied a different, otherwise uncovered date was moved to that date. Within the study window from 2018-04-01 to 2026-08-31, 129 data dates (4.2%) have no usable report. Of these, 90 are caused by the website itself: its listing points 118 report titles to only 23 files, one of which is shared by 72 report dates.

Deviation settlement rate files. Three layouts correspond to regulatory regimes, which the panel records per day:

- **R1** (2018-12-19 to 2022-12-04): frequency-linked charges with daily average exchange prices.
- **R2** (2022-12-01 to 2024-09-15): the 2022 Regulations [6], in force from 5 December 2022. Files in the new layout are published from 2022-12-01, so both layouts exist for the first days of December 2022, and the panel keeps the later listing.
- **R3** (from 2024-09-16): the 2024 Regulations [7]. The normal rate of charges for deviation is the highest of the day-ahead clearing price, the real-time clearing price and an ancillary-service-based rate.

Rates published in paise per kWh are converted to rupees per MWh. Block values are summarised per day and bid area as mean, maximum, standard deviation and the share of blocks at the regulatory cap. Where the same date appears in several files, the later listing is kept and the choice logged.

Daily generation report. Each spreadsheet is a hierarchy of region, State, sector, fuel type, station and unit rows. Column positions differ between years and are located from header text. In some files, sector and type rows appear without labels or carry subtotal values; these are recognised and skipped. The sum of station rows equals the report's own State total on every accepted date (largest absolute difference 4.5×10^{-13} GWh). 3 files with a shifted layout and no total rows cannot be validated and are excluded.

Station coal-stock days outside 0–180 are treated as entry errors. State-days containing an isolated station value above three times its capacity-based maximum, while the station's recent capacity factor is normal, are flagged. Stations are aggregated by entity and fuel (coal including lignite, gas, diesel, hydro, nuclear).

Daily renewable generation report. A workbook contains up to three tables in varying sheet order:

- renewable generation located in the State, including inter-State generating stations;
- renewable generation in the State's own control area (from 2020-06-01);
- a station list for inter-State stations.

Tables are classified by the title printed inside each sheet and checked by magnitude, because the first total can never be below the second; both definitions are released.

The date printed in the report is used. The website listing date differed from it for 10 files. Rows are accepted when wind, solar and other renewable generation add up to the total. Files that omit the "other" column (58 State-days) are recognised because wind plus solar then equals the total. Rows matching neither layout (47 and 23 State-days for the two definitions) are set to missing. One file containing the same table twice is not used. The portal files have the same layout and are read by the same parser.

Carbon intensity. Station names in the CO_2 database and in the daily generation report do not match reliably, so emission factors are built per entity, fuel and financial year. A factor is CEA absolute emissions divided by CEA net generation. Because the daily report gives gross generation, factors are multiplied by the net-to-gross ratio of each entity and fuel over the financial years in which the daily report is complete (national ratios: 0.927 for coal, 0.966 for gas).

Daily emissions are gross generation by fuel times the factor. A database version is published months after its financial year ends (version 22.0 in August 2026 [4]). The panel therefore provides factors lagged by two financial years for use as forecast inputs, and by one financial year for ex-post accounting. A consumption-based intensity adds net drawal at the day's regional generation intensity.

Weather. IMD binary grids are read as day × latitude × longitude arrays; the orientation was verified on land and sea cells and on known climatology. Entity weather is the mean over one to four load centres (capital and largest demand cities). It uses the nearest valid temperature cell and the mean of rainfall cells within ±0.25°. Cooling and heating degree days use base temperatures of 24 °C and 18 °C.

Quality control

Source tables are never altered. Values that fail a plausibility rule are set to missing in the panel and flagged:

- energy met of zero or more than five times its 29-day median (6);
- maximum demand inconsistent with energy met (16);
- energy shortage exceeding energy met (18);
- over/under-drawal exceeding energy met plus scheduled drawal (2);
- generation above capacity × 24 h with a 15% overload margin (13 State-days);
- isolated station typing errors (145 State-days).

Missing values are not imputed. The only fill values are structural zeros: a fuel with no monitored capacity in an entity on a reporting date is recorded as zero generation.

Publication timing and information sets

For each variable the field registry records the delay after which it is public. Delays were measured from the publishers' metadata:

- Power Supply Position reports appear a median 10.6 h after the end of the data day (90th percentile 36.7 h, 1,064 reports).
- Daily deviation rate files are uploaded in batches a median 8.8 days after the data date (range 4.7–71.7 days, 267 files).

The registry uses availability delays of one day for Power Supply Position and weather variables, two days for generation, renewable and carbon variables, and 14 days for prices.

Use of AI tools

During the preparation of this manuscript, the author used Claude (Anthropic) to assist with English language editing, drafting of revised text, consistency checking across tables and narrative text and formatting of the revised manuscript. All experimental design, data collection, model development, the original code implementation, and interpretation of results were performed entirely by the author. The author carefully reviewed and verified all AI-assisted output and takes full responsibility for the content of this publication.

Data Records

The files described in this section are version 1.1 of the RE-FUSED record on Zenodo [19]. The main record is `refused_state_day.parquet`, one row per entity and day from 2018-04-01 to 2026-08-31 (104,550 rows, 107 columns). It is accompanied by the following files:

- `refused_entities.csv`: entity registry;
- `FIELD_REGISTRY.csv`: source, unit, availability delay and missing share of every column;
- `QC_REPORT.md`: quality-control counts and cross-source checks;
- `DATASHEET.md`: datasheet;
- `SOURCE_REGISTRY.md`: source registry;
- `CHECKSUM.txt`: content hash;
- the interim tables at native resolution, including station-day generation (1,547,930 rows), bid-area prices, entity–fuel–year emission factors (453 cells) and entity-day weather.

Columns are grouped by prefix. `dem_` is demand, `dev_` deviation, `gen_` conventional generation and `re_` renewables. `prc_` holds prices and `co2_` carbon. `sys_`, `reg_` and `nat_` are system, regional and national context, and `wx_` is weather. `cal_`, `avail_` and `qc_` are calendar fields, source availability flags and quality flags.

Energies are in GWh (1 MU), power in MW, prices in rupees per MWh and emissions in tonnes of CO_2.

Prices begin on 2018-12-19, the first published deviation rate file. State renewable data begin on 2019-07-01 and end on 2026-08-31. IMD gridded weather ends on 31 December 2025. Conventional generation is missing for 74 days between 13 March and 31 May 2020, when no daily generation reports were published. Maximum under-drawal is printed in the source reports only from 2023, which explains most missing deviation cells before 2025.

Data Overview

Figure 2 shows, for each source, the share of entity-days with data in every month, and Table 2 the mean share of missing cells by variable block before and after 1 April 2025. Gaps follow the sources: no value is imputed (Methods), and the dates at which sources begin and end are listed under Data Records.

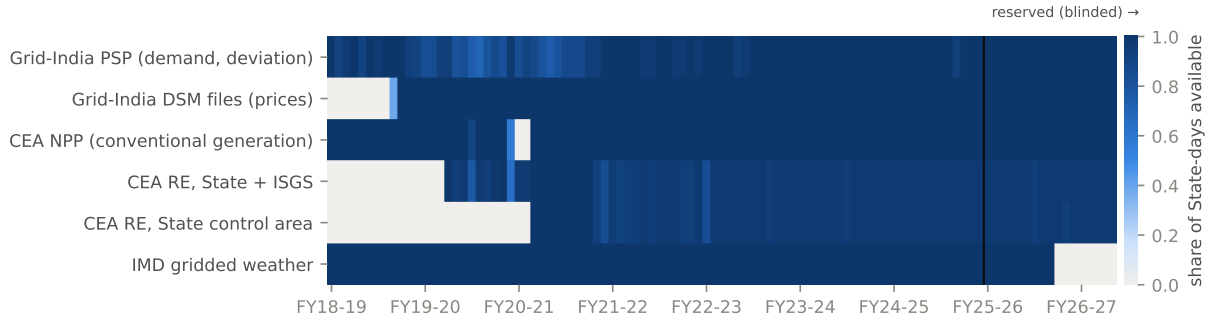


Figure 2: Share of entity-days with data, by source and month. The vertical line marks 1 April 2025, after which the panel is used as a reserved evaluation period in companion studies; this figure shows availability only.

Table 2: Mean share of missing cells by variable block, before 1 April 2025 and after.

Block	2018-04 to 2025-03 (%)	2025-04 to 2026-08 (%)
Demand	5.1	0.0
Deviation	20.9	10.3
Conventional generation	3.1	0.0
Renewables	27.4	3.2
Prices	18.5	0.0
Carbon	12.6	5.5
System stress	19.1	14.7
Regional and national	7.8	2.7
Weather	0.0	46.9

Technical Validation

Internal consistency with the sources' own totals. For every accepted daily generation report, the sum of parsed station rows reproduces the State totals printed in the same report (largest absolute difference 4.5×10^{-13} GWh over 3,019 dates). This check detected three parser failure modes during development, which were corrected:

- subtotal rows without labels;
- sector and type rows carrying values;
- a report date stored only as a date cell.

For renewable generation, the in-report arithmetic (wind + solar + other = total) holds on all accepted rows by construction, and the classification of the two table definitions is checked by magnitude on every file. The 18 reports published on both CEA listings (1 to 18 November 2025) give identical values in all 5,040 parsed cells (0 differences), so the change of portal introduces no break in the series.

Agreement with an independent copy. A previously collected aggregator copy of the same CEA reports was compared with the panel up to 31 March 2025 (Fig. 3):

- **Conventional generation:** 100.0% of 75,485 State-days agree within 1%, and 99.1% within 0.01 GWh.
- **Renewable generation** (State including inter-State stations): 99.9% of 60,619 State-days agree within 1%.

The remaining renewable disagreements trace to the copy: it assigns some reports to the website's listing date, which for 10 files differs from the date printed in the report (for example, a report for 30 September listed on 30 October 2021).

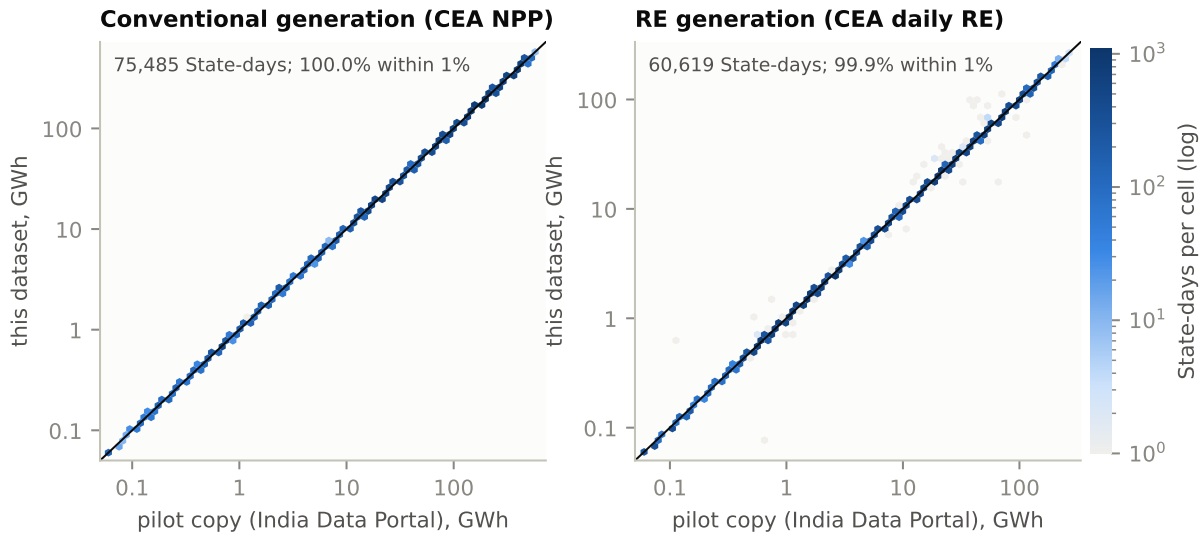


Figure 3: State-day generation in this panel against an independent aggregator copy of the same CEA reports (logarithmic axes; values up to 31 March 2025). The diagonal marks equality.

Dates. Grid-India's listing dates, the dates printed on report tables and the dates on cover letters were compared for all Power Supply Position reports. Collisions between different reports claiming the same date were resolved by the cover letter in 2 cases, and no report is used for two dates.

Physical plausibility.

- Quality-control exclusions are rare and listed individually (Methods).
- The national net-to-gross ratio of coal generation (0.927) is consistent with auxiliary consumption at coal stations. Entity-level coal emission factors lie between 0.84 and 1.04 t CO_2 per MWh of gross generation (median 0.91).
- Weather at the Delhi load centre has a mean January maximum temperature of 19.4 °C and a May mean of 39.7 °C. The grid cell containing Mumbai received 1,532 mm of rain in July 2023, confirming the grid orientation.

Publication timing. Figure 4 shows the measured delays used for the information sets. Most Power Supply Position reports appear about half a day after the end of the data day, with a tail beyond a day at weekends. Deviation rate files arrive in weekly batches.

Determinism. Rebuilding the panel from the raw files reproduces the content hash exactly. The hash is computed over row values independent of file encoding.

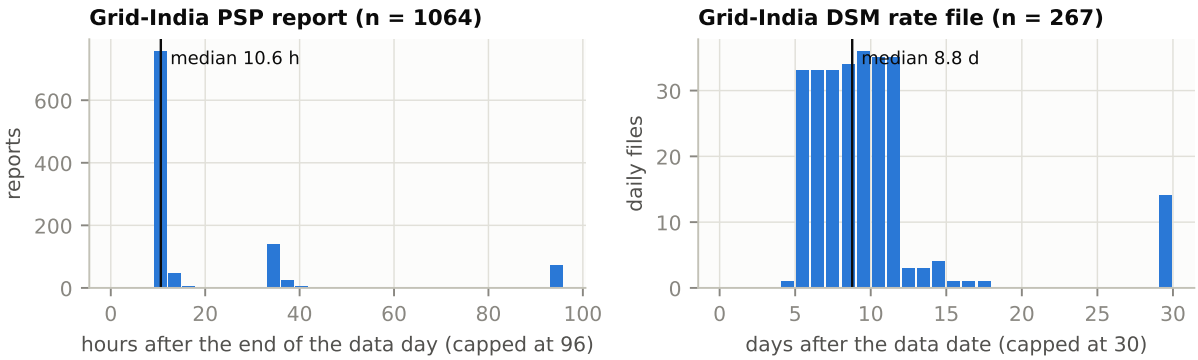


Figure 4: Publication delays measured from the publisher's file metadata: Power Supply Position reports (left, data dates from 1 April 2025) and daily deviation rate files (right, data dates from 1 January 2025).

Usage Notes

Respecting information sets. A model issuing a forecast on day τ may use a column only up to date τ minus its availability delay in the field registry. Using the latest row for every column assumes instant publication, which is not what grid operators, utilities or researchers observe. Carbon columns ending in `_acc` and IMD final gridded weather are ex-post products.

Known gaps and breaks.

- Scheduled drawal in the Power Supply Position report is the final schedule, including revisions made during the day. It is not a day-ahead schedule, and deviations measured against it are smaller than deviations from day-ahead positions would be.
- Deviations are daily net values. Over- and under-drawal within a day offset each other; maximum over- and under-drawal are the only intra-day severity indicators.
- Conventional generation is location-based and includes central stations whose output is scheduled to other States. It is not the energy supplied to the State.
- Price series change definition with the regulatory regimes R1–R3 (column `prc_regime`).
- Gridded weather ends on 31 December 2025, the last year that IMD had published. State renewable data from 19 November 2025 come from CEA's new portal (version 1.1); version 1.0 has none after 18 November 2025.

Figures in this descriptor. Companion studies use the period from 1 April 2025 as a reserved evaluation period. For that reason, the figures and value comparisons here use dates up to 31 March 2025; the coverage figure shows availability only.

Reuse terms. All source files are public reports of Government of India bodies. Their websites do not state an open licence. The release therefore consists of:

- the code and manifests needed to re-download and verify every raw file;
- the derived panel and interim tables, with attribution to Grid-India, CEA and IMD.

Raw report files are not redistributed. Users should cite the original publishers alongside this descriptor. After our downloads of September 2026 the Central Electricity Authority restructured its website: its daily renewable reports from September 2024 onwards are on the RE generation portal, but older renewable reports and the CO_2 database files were no longer at the addresses in the manifests when checked on 25 October 2026. The manifests keep each file's address and SHA-256 hash, so a copy obtained from the authority can still be checked.

Data Availability

The State-day panel is built only from reports published by Grid Controller of India, the Central Electricity Authority and the India Meteorological Department. The manifests in the record give the address, retrieval time and SHA-256 hash of every report used, so a copy obtained from the agencies can be checked; some older Central Electricity Authority files are no longer at the addresses recorded in September 2026, because the authority has since moved its website. The panel described here is version 1.1.0 of the RE-FUSED record on Zenodo [19], at `https://doi.org/10.5281/zenodo.23301897` (this DOI always resolves to the latest version), with the panel and documentation under the CC BY 4.0 licence and the code under the MIT licence. It holds the State-day panel (`refused_state_day.parquet`), the entity and field registries, the quality-control report, the datasheet, the source registry, the content checksum, the interim tables at native resolution, the manifests of every downloaded file and the code that downloads, parses, builds and validates the panel. Version 1.0.0 (`https://doi.org/10.5281/zenodo.23301898`) is the earlier, frozen panel used by the companion studies; it differs only in the renewable columns from 19 November 2025 onwards.

Code Availability

The acquisition, parsing, build and validation code is part of the release (Python 3.10). The steps are:

- `acquire/` downloads the files and writes the manifests;
- `b01`–`b06` parse each source into interim tables;
- `b07` builds the panel, field registry, quality-control report and checksum;
- `b08` recomputes every validation statistic reported here.

The numbers in this descriptor are generated from the validation output, and its figures from the panel, by two scripts included in the release. A leakage test checks that features built with the registry's availability delays do not change when unpublished values are perturbed.

The code, the panel and its documentation are released together on Zenodo, and the DOI of the version described here is `https://doi.org/10.5281/zenodo.23301897` (it always resolves to the latest version). The code and build logs are also openly available at `https://github.com/Ohi-Me/RE-FUSED`.

References

[1] Grid Controller of India Limited. Daily Power Supply Position Reports, National Load Despatch Centre. Daily PSP Report page of grid-india.in, files listed through the site's public file interface; reports 2017–2026, accessed 14 September 2026, 2026. URL `https://grid-india.in/en/reports/daily-psp-report`.

[2] Central Electricity Authority. Daily Generation Report, National Power Portal. Published reports of the National Power Portal, daily sub-report 2 (public-reports/cea/daily/dgr/DD-MM-YYYY/dgr2-YYYY-MM-DD.xls); 2018–2026, accessed 14 September 2026, 2026. URL `https://npp.gov.in/publishedReports`.

[3] Central Electricity Authority. Daily Renewable Generation Report, Renewable Project Monitoring Division. Reports to 18 November 2025 from the report listing of cea.nic.in, accessed 14 September 2026; reports from November 2025 from CEA's RE generation portal, accessed 25 October 2026, 2026. URL `https://gen-re.cea.gov.in/reports`.

[4] Central Electricity Authority. CO_2 baseline database for the Indian power sector, user guide version 22.0. Ministry of Power, Government of India; database files downloaded on 14 September 2026 from the CDM CO_2 baseline database page of cea.nic.in, which has since

moved; every file URL and hash is in the release manifest, August 2026. URL `https://cea.nic.in`.

[5] Grid Controller of India Limited. Deviation Settlement Mechanism Rate Files, National Load Despatch Centre. DSM Rates page of grid-india.in; files 2018–2026, accessed 14 September 2026, 2026. URL `https://grid-india.in/en/markets/dsmrates`.

[6] Central Electricity Regulatory Commission. Central Electricity Regulatory Commission (Deviation Settlement Mechanism and Related Matters) Regulations, 2022. Notification No. L-1/260/2021/CERC, dated 14 March 2022, 2022. URL `https://www.cercind.gov.in/Regulations/168_reg.pdf`.

[7] Central Electricity Regulatory Commission. Central Electricity Regulatory Commission (Deviation Settlement Mechanism and Related Matters) Regulations, 2024. Notification No. L-1/260/2021/CERC, dated 5 August 2024, 2024. URL `https://www.cercind.gov.in/regulations/192-Noti.pdf`.

[8] D. S. Pai, Latha Sridhar, M. Rajeevan, O. P. Sreejith, N. S. Satbhai, and B. Mukhopadhyay. Development of a new high spatial resolution ($0.25^\circ \times 0.25^\circ$) long period (1901–2010) daily gridded rainfall data set over India and its comparison with existing data sets over the region. *MAUSAM*, 65(1):1–18, 2014. doi: 10.54302/mausam.v65i1.851.

[9] A. K. Srivastava, M. Rajeevan, and S. R. Kshirsagar. Development of a high resolution daily gridded temperature data set (1969–2005) for the Indian region. *Atmospheric Science Letters*, 10(4):249–254, 2009. doi: 10.1002/asl.232.

[10] India Meteorological Department. Gridded daily rainfall (0.25°) and maximum/minimum temperature (1°) data, Climate Monitoring and Prediction Group, Pune. Yearly binary files 2017–2025, accessed 14 September 2026, 2026. URL `https://imdpune.gov.in/cmpg/Griddata/Rainfall_25_Bin.html`.

[11] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021. doi: 10.1145/3458723.

[12] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, et al. The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3:160018, 2016. doi: 10.1038/sdata.2016.18.

[13] Frauke Wiese, Ingmar Schlecht, Wolf-Dieter Bunke, Clemens Gerbaulet, Lion Hirth, Martin Jahn, et al. Open power system data – frictionless data for electricity system modelling. *Applied Energy*, 236:401–409, 2019. doi: 10.1016/j.apenergy.2018.11.097.

[14] Lion Hirth, Jonathan Mühlenpfordt, and Marisa Bulkeley. The ENTSO-E transparency platform – a review of Europe's most ambitious electricity data platform. *Applied Energy*, 225:1054–1067, 2018. doi: 10.1016/j.apenergy.2018.04.048.

[15] Stefan Pfenninger and Iain Staffell. Long-term patterns of European PV output using 30 years of validated hourly reanalysis and satellite data. *Energy*, 114:1251–1265, 2016. doi: 10.1016/j.energy.2016.08.060.

[16] Iain Staffell and Stefan Pfenninger. Using bias-corrected reanalysis to simulate current and future wind power output. *Energy*, 114:1224–1239, 2016. doi: 10.1016/j.energy.2016.08.068.

[17] Debani Prasad Mishra, Sarita Samal, Rohit Kumar, Arun Kumar Sahoo, and Surender Reddy Salkuti. Renewable energy optimization for sustainable power generation. *International Journal of Informatics and Communication Technology*, 15(1):365–373, 2026. doi: 10.11591/ijict.v15i1.pp365-373.

[18] Stefan Pfenninger, Joseph DeCarolis, Lion Hirth, Sylvain Quoilin, and Iain Staffell. The importance of open data and software: Is energy research lagging behind? *Energy Policy*, 101:211–215, 2017. doi: 10.1016/j.enpol.2016.11.046.

[19] Rohit Kumar. RE-FUSED: State-day panel of India's power system from official data, with forecasting, forecast correction and scheduling code, results and logs. Version 1.1.0, Dataset, Zenodo, 2026. URL `https://doi.org/10.5281/zenodo.23301897`.

Funding

This research did not receive a specific grant from any funding agency in the public, commercial or not-for-profit sectors.